

How Much Adaptation Is Enough?

Drift Geometry and Minimal Calibration in Brain-to-Text BCI

Anonymous Authors

Abstract

Brain-to-text BCI decoders must remain usable as neural signals change across recording sessions, but daily recalibration is costly for users and requires labeled data. We audit how much input-side adaptation is needed when the official T15 brain-to-text GRU is evaluated across sessions with its backbone and output head frozen. Native day-specific input layers reproduce low validation phoneme error (9.06% PER), whereas forcing sessions through non-native input layers increases PER to 22.67–34.43%. Unsupervised mean/variance corrections fail to recover this loss, and a supervised input-layer adapter trained from 20 labeled calibration trials improves the fixed-middle stress setting only from 22.67% to 18.59% PER. The strongest low-cost result is source-layer management: at $K = 20$, immediately previous-session reuse reaches 12.21% PER, while short-window covariance-based retrieval of a past input layer reaches 12.74% PER and exposes older-state candidates. However, geometry is not yet a standalone deployment policy: when it selects an older non-previous source at $K = 20$, that source beats previous in only 3 of 11 cases. These results support a recency-aware adaptation ladder: reuse the previous input state when it is recent, use drift geometry to nominate older candidates when recency becomes unreliable, and reserve labeled calibration for cases where no stored input state is trustworthy.

1 Introduction

Do we really need to recalibrate or retrain a brain-to-text decoder every day? For a deployed BCI, this is a practical rather than theoretical question: daily calibration takes user time, requires labeled data, and may still be unnecessary when a compatible input state already exists. Long-term stability has long been recognized as a central challenge for practical BCI deployment [Vaadia

and Birbaumer, 2009]; here, “drift” means that the neural feature distribution recorded on a new day no longer matches the distribution seen by the decoder during earlier sessions. We show that the answer is not simply yes or no. In T15, the shared GRU decoder can remain frozen, but the input state must be chosen carefully. Previous-session reuse is the strongest simple default; short-window geometry can nominate older candidates; and labeled calibration is best treated as an escalation step rather than the first response.

We test this input-state view in the official T15 brain-to-text GRU baseline from a rapidly calibrating speech neuroprosthesis study [Card et al., 2024]. The model contains learned day-specific input layers, followed by a shared GRU and output head. Each input layer acts like a small session-specific adapter that maps that day’s neural features into the representation expected by the shared GRU decoder. This structure creates a useful stress test: if a target session is decoded through the wrong input layer, performance measures the decoder-facing cost of cross-session input mismatch. We then compare simple feature corrections, small learned adapters, and geometry-guided source selection as rungs of a lightweight adaptation ladder.

The contribution is not a new speech decoder. Instead, we provide a decoder-facing adaptation audit:

- Wrong input states break the decoder: native-day PER is 9.06%, while fixed non-native input layers increase PER to 22.67–34.43%.
- Simple mean/scale correction is insufficient; input-layer calibration with $K = 20$ improves PER to 18.59%, but remains far from native-day decoding.
- Previous-session reuse is the strongest simple policy at $K = 20$ (12.21% PER), while geometry selection is close (12.74%) and reveals older-state candidates.
- Geometry is a retrieval signal, not a standalone policy: at $K = 20$ it selects older states in 11 sessions, but wins only 3 times, motivating a confidence gate.

We do not claim that neural drift geometry is new, or that covariance distance alone solves cross-session adaptation. Rather, we show that in a deployed-style brain-to-text decoder with learned day-specific input lay-

ers, short-window geometry is actionable as a candidate-retrieval signal and as a way to make recency an explicit adaptation prior.

Figure 1 summarizes this adaptation-ladder view: keep the shared decoder frozen, estimate the new input state from a short beginning-of-day window, and choose the lightest input-side action supported by the evidence.

2 Related Work

Brain-to-text BCIs have progressed from handwriting-based decoding to high-performance speech neuroprostheses, but long-term use still requires session-specific adaptation or recalibration as neural recordings change over time [Willett et al., 2021; Willett et al., 2023; Card et al., 2024]. Cross-session stability has also been studied through neural geometry and manifold alignment: prior work shows that low-dimensional neural structure can be aligned across recording instabilities, and dynamics-aware methods can stabilize BCIs over longer timescales [Degenhart et al., 2020; Karpowicz et al., 2025]. Our setting is narrower: rather than learning a new invariant decoder, we ask whether short-window geometry can help choose among existing day-specific input states. Continual recalibration and session-invariant learning provide complementary ways to update or regularize decoders [Fan et al., 2023; Zhang et al., 2026]; here, we audit low-cost actions available before retraining: previous-session reuse, geometry-guided source retrieval, and lightweight input-layer calibration.

3 Data and Decoder

T15. The main experiments use 41 T15 validation sessions from the official brain-to-text copy-task dataset [Card et al., 2024]. Neural features are passed through the official GRU checkpoint and decoded greedily at the phoneme level. We compute phoneme error rate (PER) after standard CTC collapse; lower PER is better.

Official baseline. The decoder contains learned day-specific input layers, a shared GRU backbone, and an output layer over phoneme classes. Each input layer is a small learned linear mapping that projects that day’s neural features into the representation expected by the shared GRU. In this paper, an input state refers to the stored input-layer weights associated with a particular session. Native-day decoding uses each session’s own input state. Cross-day stress tests use a fixed or selected non-native source input state while keeping the GRU and output head unchanged.

Beginning-of-day setting. We use the first $K \in \{5, 10, 20\}$ trials of a target session for geometry estimation or supervised input calibration, and evaluate PER only on the remaining trials. This intentionally simulates a beginning-of-day operating regime rather than random access to the whole session. T12 and inner-speech analyses are used only as geometry-only supporting checks, not decoder replications.

4 Methods

4.1 Decoder-Facing Metrics

For each T15 validation trial, we compute greedy phoneme predictions from the CTC output stream and apply CTC collapse before computing phoneme error rate. For a set of trials $\mathcal{D}$, the reported PER is

$$\text{PER}(\mathcal{D}) = \frac{\sum_{i \in \mathcal{D}} \text{edit}(y_i, \hat{y}_i)}{\sum_{i \in \mathcal{D}} |y_i|}.$$

Unless stated otherwise, table-level PER values are phoneme-weighted aggregate PER over the relevant evaluation trials. Per-session counts are used only for state-ments such as how often a candidate source wins, loses, or ties. We avoid mixing PER with derived recovery percentages in the main result tables: lower PER is always better, and improvements are reported as direct PER changes.

4.2 Adaptation and Source Selection

We compare native-day decoding against three fixed non-native source layers: early (t15.2023.08.13), middle (t15.2023.11.26), and late (t15.2025.04.13). These sources are chosen to span the longitudinal recording period rather than to optimize fixed-source performance. Using the middle source layer, we test a ladder of increasingly expensive input-side interventions: no correction, unsupervised mean/scale moment matching, supervised diagonal affine calibration, and supervised input-layer calibration. Simple scalar normalizations, including target z-scoring and source moment matching, test whether channel-wise mean and variance alignment is sufficient to explain the cross-session mismatch. Learned adapters use the first K labeled trials and are evaluated on the remaining trials. The GRU backbone and output head remain frozen; only the input transform is learned using CTC loss.

Neural feature geometry. The neural data for each session can be viewed as a tensor of trials $\times$ time bins $\times$ neural features. For the geometry analysis, we collapse the trial and time dimensions and treat each time bin as one neural-feature vector in channel space. Thus, a session becomes a cloud of points whose dimensions are the neural channels/features. The covariance matrix summarizes the shape of this cloud by measuring how neural features co-vary across all time bins and trials. Intuitively, two sessions are geometrically close if these population-activity clouds have similar covariance or subspace structure, even if they occur on different days. For K-shot source selection, the first K unlabeled target trials are used to estimate the target covariance. We use relative Frobenius covariance distance:

$$d_{\text{cov}}(t, s) = \frac{\|\text{Cov}(S_t^{(K)}) - \text{Cov}(S_s)\|_F}{\|\text{Cov}(S_s)\|_F + \epsilon}.$$

Here $\epsilon = 10^{-8}$ is a small stabilizer to prevent division by zero. This source-normalized distance is used as a simple scale-aware diagnostic rather than as a claim of optimal

Recency-aware adaptation ladder for cross-session BCI drift

Estimate new-session geometry → retrieve a compatible input state → decode with the GRU frozen

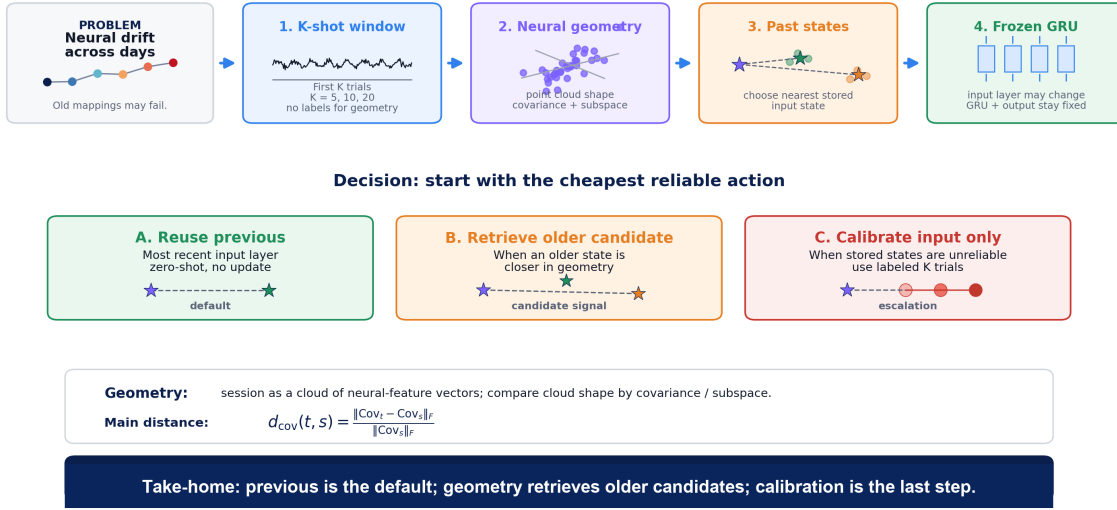


Figure 1: Recency-aware adaptation ladder. A short beginning-of-day window estimates target-session geometry, compares it to stored input states, and chooses the lightest input-side action: reuse previous, retrieve an older candidate, or calibrate the input layer only. The GRU backbone and output head remain frozen.

geometry matching. We then compare this distance to each past source session and reuse the input layer of the closest past session:

$$s^*(t) = \arg \min_{s \in \mathcal{S}_{\text{past}}(t)} d_{cov}(t, s).$$

For subspace diagnostics, we also compute PCA-based subspace distances between each target–source pair. We form low-dimensional bases from the top 20 principal components of the target and source neural-feature matrices, then compute the principal angles between these two subspaces. The reported mean principal angle is the average of these angles in degrees; smaller values indicate more similar dominant neural activity subspaces. We compare this geometry policy to previous-session reuse, $s_{\text{prev}}(t)$, because recency is the strongest prior available in a longitudinal BCI. We also compare bounded input-state libraries: previous only, last 3 sessions, last 5 sessions, and all past sessions.

5 Experiments

We first audit decoder-facing adaptation on T15, then use T12 and inner-speech blocks as geometry-only supporting checks.

Before presenting the detailed experiments, Table 1 summarizes the main empirical story. The results follow a simple adaptation ladder: wrong input states strongly degrade decoding, simple moment correction fails, small supervised calibration helps only partially, and the strongest low-cost intervention is to reuse or retrieve a compatible past input state. Thus, the central question becomes not whether adaptation is possi-

Table 1: Main findings at a glance. Lower PER is better.

Question	Result	Interpretation
Does the native input state matter?	Native-day: 9.06% PER	Correct day-specific input layers work well.
What if the input state is wrong?	Fixed non-native: 22.67–34.43% PER	Cross-session mismatch strongly degrades decoding.
Is mean/scale correction enough?	Channel-wise moment match: 22.73% PER	Simple distribution matching fails.
Does small calibration solve it?	Input layer, $K = 20$: 18.59% PER	Calibration helps, but remains far from native-day.
What is the best low-cost policy?	Previous source, $K = 20$: 12.21% PER	Recency is the strongest default.
Does geometry help?	Geometry, $K = 20$: 12.74% PER; older wins 3/11	Geometry retrieves candidates, but needs a gate.

ble, but which input state should be used before any supervised calibration is attempted.

5.1 Cross-Day Stress Test and Adaptation Ladder

Finding 1: the day-specific input layer is essential. Native-day decoding reaches 9.06% phoneme-weighted PER, confirming that the local evaluation reproduces the expected behavior of the official checkpoint. When all sessions are forced through non-native source input layers, PER rises to 22.67–34.43%, harming 40/41 sessions for each source choice.

The natural first hypothesis is that cross-session mismatch is mostly a mean/variance shift. The data do

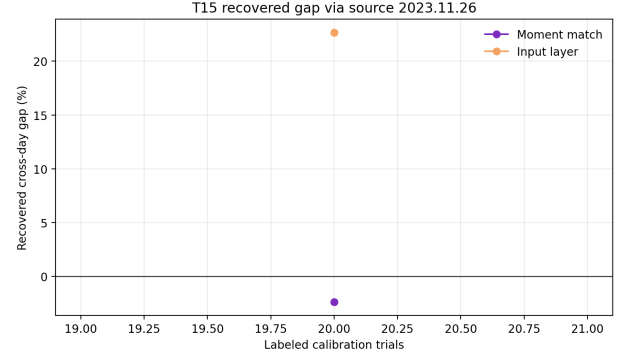
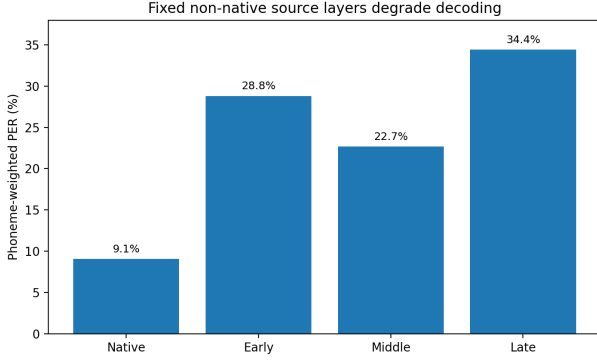


Figure 2: Left: fixed non-native source layers strongly degrade T15 phoneme decoding relative to native-day decoding. Right: small learned calibration improves over the fixed-middle source but remains far from native-day decoding.

Table 2: T15 decoder-facing stress test and adaptation ladder. Aggregate, best-session, and worst-session values are phoneme-weighted PER. Fixed-source and moment-correction rows use the middle source t15.2023.11.26. Learned adapter rows use $K = 20$ calibration trials and are evaluated on remaining trials. Lower PER is better.

Condition	Aggregate PER ↓	Best session	Worst session
Native-day	9.06%	0.80% (11-19)	23.82% (03-14)
Fixed early	28.81%	7.38% (08-13)	58.28% (03-14)
Fixed middle	22.67%	1.20% (11-19)	56.44% (03-30)
Fixed late	34.43%	16.40% (01-12)	49.04% (07-19)
Channel-wise moment match	22.73%	1.20% (11-19)	56.90% (03-30)
Diag. affine, $K = 20$	20.21%	1.94% (11-26)	53.26% (03-30)
Input layer, $K = 20$	18.59%	2.46% (11-26)	52.23% (03-30)

Table 3: Beginning-of-day source selection on remaining validation trials. Previous-session recency is the strongest simple baseline, while K-shot geometry selection remains close and often selects non-previous sources. Lower PER is better.

Method	$K = 5$	$K = 10$	$K = 20$
Native-day	8.95%	8.88%	8.90%
Fixed middle	22.51%	22.40%	22.30%
Previous source	13.09%	12.84%	12.21%
K-shot geometry source	14.00%	13.19%	12.74%
Best simple override	13.09%	12.90%	12.18%

Table 4: Previous-session recency horizon. Previous-session reuse is highly effective when the stored state is recent, but degrades when the previous available state is stale. These values motivate geometry-based candidate retrieval; they do not by themselves prove that covariance-only retrieval is a safe override policy.

Age of previous input state	Previous-source PER	Operational reading
0–3 days	9.97%	Reuse previous is often sufficient.
4–14 days	13.75%	Recency remains useful but weaker.
>14 days	24.54%	Blind reuse becomes risky.

not support this: moment matching remains essentially tied with no correction. Supervised diagonal affine and input-layer calibration improve PER beyond moment matching, with input-layer calibration reaching 18.59% at $K = 20$. However, this remains far from 9.06% native-day decoding, so small calibration is useful but does not replace a well-matched prior input state.

5.2 T15 Geometry and Beginning-of-Day Source Selection

Finding 2: T15 drift is geometrically structured over time. Before using geometry for retrieval, we first verify that T15 sessions have temporally structured input geometry. Figure 3 shows that pairwise covariance shift increases with temporal separation, while nearby sessions form a diagonal band of similar geometry. This supports recency as a strong prior, but the map is not purely chronological.

The beginning-of-day version is more realistic: the first K trials estimate geometry or train a small adapter, and evaluation is performed only on the remaining trials. Past-only source selection reaches 14.00%, 13.19%, and 12.74% PER at $K = 5, 10, 20$, far below fixed-middle decoding on the same subsets. However, the immediately previous input layer is even stronger: 13.09%, 12.84%, and 12.21% PER. This result should be read as a strong recency baseline rather than as a failure of geometry: nearby days are usually close in geometry, so previous-

session reuse already captures much of the useful structure.

Table 4 makes the recency horizon explicit. Previous-source PER is 9.97% when the last available input state is 0–3 days old, 13.75% at 4–14 days, and 24.54% beyond two weeks. Thus, previous is a strong default only when it is truly recent. When the previous input state is stale, geometry becomes most relevant as a retrieval signal for candidate input states, but a deployment policy still needs a confidence gate before overriding recency.

5.3 Geometry Retrieves Older Candidates but Needs a Gate

Finding 3: geometry retrieves older candidates, but does not reliably replace recency. The previous-session input

T15 longitudinal geometry: days close in time are usually close in input-state geometry

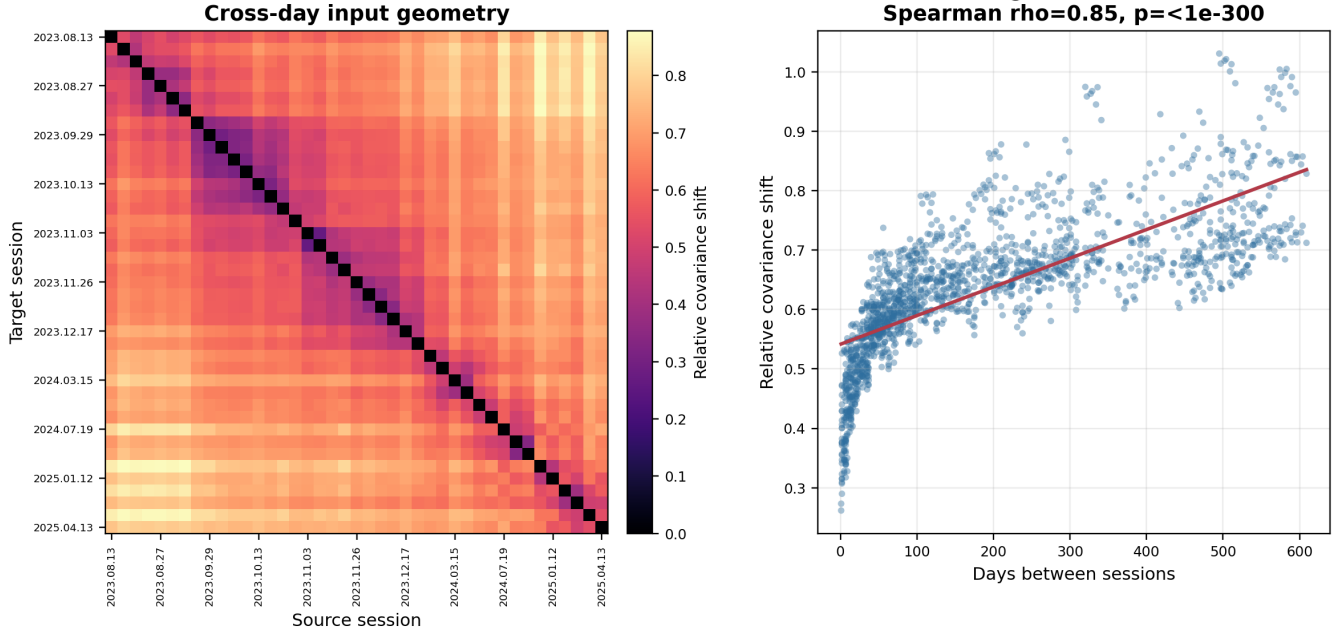


Figure 3: T15 longitudinal geometry map. Left: pairwise relative covariance shift between source and target sessions, ordered by date, shows a diagonal band of nearby sessions with similar input geometry. Right: covariance shift increases with temporal separation across source-target pairs (Spearman $\rho = 0.85$), indicating that session geometry drifts with time while retaining non-monotonic structure.

layer is the strongest simple baseline, but it is not always the geometrically nearest source. At $K = 20$, geometry selects the previous source in 26 sessions and an older non-previous source in 11 sessions. Among those 11 older selections, the older source achieves lower PER in 3 cases, the previous source remains better in 7 cases, and 1 case is tied. When geometry incorrectly selects an older source, the mean penalty relative to previous-session reuse is 3.23 percentage points (median 3.49, max 4.55). Thus geometry identifies plausible older-state candidates, but covariance distance alone is not sufficient as a deployment policy: closer in geometry does not always mean lower PER.

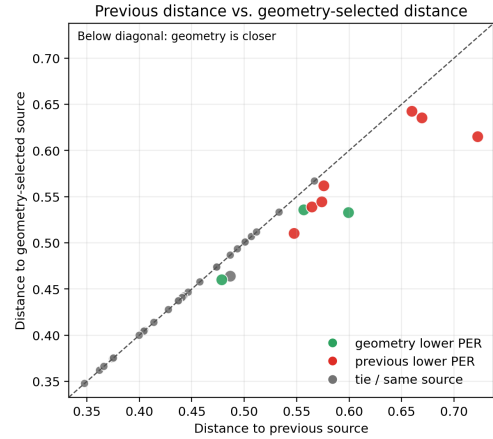


Figure 4: Geometry retrieves closer candidates but is not a standalone policy. Points below the diagonal have a geometry-selected source that is closer than the previous source in covariance distance. Green points indicate cases where geometry achieves lower PER; red points indicate cases where previous remains better.

This motivates retaining a compact memory of input states rather than only the latest layer. We restrict K -shot geometry selection to previous only, the last 3 sessions, the last 5 sessions, or all past sessions. The immediately previous session remains the best aggregate policy, but a small recent library nearly matches all-past

Table 5: K-shot input-state library-size ablation. PER is measured on remaining validation trials after using the first K trials to estimate target geometry. A bounded recent library nearly matches all-past selection, while previous-only recency remains the strongest aggregate baseline.

K	Previous	Last 3	Last 5	All past
5	13.09%	13.47%	13.55%	14.00%
10	12.84%	13.19%	13.19%	13.19%
20	12.21%	12.76%	12.74%	12.74%

Table 6: Preliminary drift fingerprints for non-previous source choices. Negative principal-angle change means the older source is closer to the target subspace than the previous source.

K	Case	n	Δ principal angle
5	older wins	3	-1.37°
5	older loses	8	-0.06°
10	older wins	4	-1.93°
10	older loses	5	+1.02°
20	older wins	3	-2.91°
20	older loses	8	+0.55°

selection.

A simple distance-ratio gate does not yet solve this problem. At $K = 20$, the best simple override improves only marginally from 12.21% to 12.18% PER, and a leave-one-session-out decision stump improves only to 12.15% while performing worse at smaller K . Older sources beat previous in only a minority of non-previous selections (3/11, 4/9, and 3/11 for $K = 5, 10, 20$), so future gates need richer confidence features.

To ask why older sources sometimes help, we compare drift fingerprints for target-previous and target-older-source pairs. Although older-source wins are rare, they are better explained by subspace geometry than by mean/scale shifts alone. When older sources win, they reduce the mean principal angle relative to previous by 1.37°, 1.93°, and 2.91° for $K = 5, 10, 20$; when older sources lose, the same subspace advantage is absent.

5.4 Geometry-Only Supporting Checks

T12 diagnostic blocks provide a secondary geometry-only check. Across spike-power and threshold-crossing feature variants, covariance distance increases with temporal separation, and geometry sometimes selects older non-previous sessions. For spike-power features, geometry selects an older source in 7/15 targets, including a 53-day non-recent match.

As a separate behavior-state check, we analyzed inner-speech behavior blocks without training a decoder or attempting inner-speech decoding. Covariance geometry retrieves matching behavior well above chance, with binned threshold crossings during the go epoch reaching 96.15% nearest-behavior accuracy across five behavior classes, compared with a 20% chance level. These results support the broader input-state view, but remain

geometry-only.

6 Discussion

The main lesson is that cross-session adaptation should not be treated as a single fixed correction. Native T15 input layers are highly effective, while fixed non-native input layers cause large decoder-facing degradation. Simple moment alignment does not emulate those layers, and learned input calibration improves PER only partially. The most useful low-cost intervention is therefore not a new adapter, but a better choice of an existing input layer.

Table 7 summarizes the deployment interpretation of the adaptation ladder.

A small number of difficult sessions, especially 2025-03-30 and 2025-03-14, dominate worst-case PER. However, sensitivity checks excluding the worst one or two sessions preserve the same qualitative ordering: fixed non-native layers remain degraded, moment matching remains ineffective, input-layer calibration remains partial, and previous/geometry source policies remain the strongest low-cost choices. We therefore retain all sessions and treat these days as stress cases for future session-quality gating.

The resulting adaptation ladder is sequential. First, reuse the previous input state when it is recent and geometrically plausible, since previous-session reuse is the strongest simple default. Second, if the beginning-of-day geometry suggests that an older stored state is closer to the target session, retrieve that older input state as a candidate for a gate rather than blindly switching to it. Third, if no stored state appears reliable and labels are available, perform lightweight residual calibration of the input layer only. This makes calibration an escalation step, not the default response to every new day.

Geometry should not be treated as a replacement for recency. At $K = 20$, it identifies older non-previous candidates in 11 sessions, but those candidates beat previous in only 3 cases. The drift-fingerprint analysis suggests that subspace mismatch is more informative than covariance distance alone when deciding whether an older state is worth retrieving. Future gates should combine subspace features, distance margins, temporal gap, decoder confidence, and session-quality diagnostics.

This framing complements stronger decoder-backbone approaches. Pretrained speech models or language-model decoding may improve absolute performance, while input-state selection addresses a different deployment question: how to keep a fixed decoder usable across sessions with minimal calibration. Future work should test this ladder with WER, language-model decoding, closed-loop evaluation, additional datasets, and session-level uncertainty estimates. Preliminary alignment probes also suggest caution: geometry-based feature transformations such as covariance alignment may introduce harms, so geometry is currently more reliable as a retrieval signal than as an automatic feature transformation.

Table 7: Operational interpretation of the adaptation ladder. PER values are evidence from this audit rather than guaranteed deployment performance. Lower PER is better.

Scenario	Naive action	Recommended ladder action	Evidence in this audit	Operational insight
Recent previous state (0–3 days)	Reuse latest state	Reuse previous (zero-shot)	Previous source: 9.97% PER	Recency wins when the last input state is truly recent; no labels or calibration are needed.
Stale previous state (>14 days)	Reuse latest state anyway	Check geometry and nominate stored candidates	Previous degrades to 24.54% PER; geometry exposes non-previous candidates	The previous layer has a time horizon; geometry should be used as a candidate signal, not as an automatic override.
No reliable stored state	Full retraining or blind reuse	Residual input calibration ($K = 20$ labeled trials)	Fixed middle: 22.67% → 18.59% PER	Calibration helps, but remains far from native-day; use it as an escalation step.
Simple shift	distribution	Mean/scale correction	Channel-wise moment match: 22.73% PER	Mean/variance alignment does not capture the decoder-relevant drift.
Geometry-selected older candidate	Always override previous	Use a confidence gate	Older selected in 11 sessions; wins in 3; false-positive cost 3.23 pp	Geometry retrieves plausible candidates, but covariance distance alone is not reliable enough as a standalone policy.

Beyond motor and speech BCI, a further direction is to test whether the input-state library view extends to cognitive or memory-related neural recordings. Recent human hippocampal and entorhinal single-neuron work shows that neural populations actively encode the temporal structure of experience [Tacikowski et al., 2024]. Such datasets could test whether geometry-guided state retrieval is specific to speech BCI drift, or whether it reflects a broader, domain-general principle of reusable neural state organization.

7 Limitations

This study is a decoder-facing audit, not a new decoding architecture. The main decoder-facing evidence comes from T15 validation sessions, so the policy conclusions should be tested on additional participants and datasets before deployment claims are made. We report greedy phoneme error rate rather than full WER or language-model decoding. T12 and inner-speech analyses are geometry-only and should not be interpreted as decoder replications. The geometry policy uses a simple covariance distance; symmetric metrics, shrinkage covariance, subspace-aware gates, and decoder-confidence features may be more reliable. A practical input-state library also requires memory management and efficient indexing. In our offline setting, source-session covariance statistics can be precomputed, so beginning-of-day retrieval only requires comparing the target covariance to stored session summaries. A deployed system may still need to prune, cluster, or compress stored input states over months or years rather than retaining every past session. The learned-adapter and K-shot source-selection experiments simulate a beginning-of-day window within the validation split and evaluate only on remaining trials. Larger windows and random-window robustness checks can be informative, but we focus on $K \leq 20$ as the minimal-calibration regime.

8 Conclusion

For T15 brain-to-text decoding, fixed non-native input layers break the decoder, moment correction is too weak, and learned input adaptation improves PER only partially. The largest practical improvement comes from selecting a compatible past input layer, with immediately previous-session reuse as the strongest simple default. Drift geometry is still valuable, but as a measurable candidate-retrieval signal rather than a standalone replacement for recency. This supports a recency-aware, geometry-guided adaptation ladder: start with the previous input state, measure drift geometry from a short unlabeled window, nominate older candidates only when recency becomes unreliable, and move to labeled calibration only when simple source selection is not enough.

Ethical Statement

This work uses publicly released intracortical BCI datasets and reports only aggregate analyses. It does not attempt to infer private inner speech or deploy a decoder. Future extensions to inner-speech or cross-behavior state selection should include explicit intent-gating and privacy safeguards.

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